

MINI PROJECT REPORT

Enterprise Student Record and Academic Management Platform

Design Thinking and Product Planning

Prepared by :

Name: Gaikwad Gaytri Anil

Class / Division: DSY AIML(B)

Academic Year: 2026-27

1. Project Overview

The Enterprise Student Record and Academic Management Platform is a proposed software solution for managing student records and academic information in a higher-education institution. The project follows a Design Thinking approach before coding, with emphasis on understanding the academic domain, stakeholders, user needs, requirements, product vision, MVP and roadmap.

The project scope is based on the supplied Unit-1 mini-project brief. The brief requires students to study admissions, course registration, attendance, examinations and results, identify manual-record problems, and prepare planning documents before development.

Source basis: supplied Unit-1 project brief. filecite turn0file0 L5-L11

2. Problem Statement

Many academic activities involve maintaining student information, attendance, examination records and academic reports. When these records are handled manually or through disconnected systems, staff may spend additional time searching, updating and preparing reports. Students may also have difficulty accessing accurate academic information.

Problem statement: Design a secure, usable and reliable platform that centralizes student academic records and supports registration, attendance, examination/result management and report generation.

3. Objectives

Understand the academic management process.

Identify stakeholders and their needs.

Understand problems in manual record management.

Apply the five Design Thinking stages: Empathize, Define, Ideate, Prototype and Test.

Prepare functional and non-functional requirements.

Define the product vision and Minimum Viable Product (MVP).

Create an initial system architecture and project roadmap.

Prepare the project structure and Git repository for later Agile development.

4. Scope of the Project

Module	Main Activities
Student Registration	Add, update and view student information
Course Registration	Maintain course/subject registration details
Attendance	Record and view attendance
Examination	Maintain examination-related academic records
Results	Generate and view academic results
Reports	Generate student and academic reports
User Management	Provide role-based access for users
Security	Protect academic information and restrict unauthorized access

5. Business Understanding

The academic workflow considered in this project is:

Student admission / registration

Course or subject registration

Attendance recording

Examination scheduling and record management

Result entry and generation

Academic report generation

Record review and maintenance

The supplied brief specifically asks students to study admissions, course registration, attendance, examination and result management, and identify issues in manual record management. filecite turn0file0 L14-L18

6. Design Thinking Process

Stage	Activities for this Project	Output
Empathize	Interview/observe students, faculty and administrative staff; understand pain points.	User observations and needs
Define	Convert observations into clear problem statements.	Problem statement
Ideate	Generate possible solutions and prioritize useful features.	Feature ideas
Prototype	Create wireframes/mock screens for key modules.	Low-fidelity prototype
Test	Show prototype to users and collect feedback.	Feedback and improvements

The supplied brief explicitly requires the five stages: Empathize, Define, Ideate, Prototype and Test.
filecite turn0file0 L19-L21

7. IBM Design Thinking

- The project applies the three principles mentioned in the brief:
 - Focus on User Outcomes – design features around useful outcomes for students, faculty and administrators.
 - Restless Reinvention – continuously improve the proposed solution based on feedback.
 - Diverse Empowered Teams – involve different stakeholders and perspectives.
- Observe → Reflect → Make cycle:

Observe	Reflect	Make
Understand user activities and difficulties.	Identify important problems and opportunities.	Create or improve a feature/prototype and test it.

These IBM Design Thinking principles and the Observe → Reflect → Make cycle are required by the supplied project brief. filecite turn0file0 L22-L25

8. Stakeholder Analysis

Stakeholder	Role / Responsibility	Main Need
Students	Use academic services and view records.	Easy access to registration, attendance, results and reports.
Faculty Members	Manage attendance and academic information.	Fast and accurate record entry.
Academic Administrators	Manage student/course records.	Centralized and reliable data.
Examination Cell	Manage examination and result records.	Accurate result processing and reporting.
IT Support	Maintain system and technical support.	Security, reliability and maintainability

The stakeholder groups above are directly based on the supplied brief. filecite turn0file0 L26-L28

9. User Personas

Persona	Goals	Responsibilities	Challenges	Expectations
Student	Access accurate academic information.	Registration, viewing attendance/results.	Finding information and tracking records.	Simple, quick and mobile-friendly access.
Faculty	Maintain academic records efficiently.	Attendance and academic record entry.	Repeated manual data entry.	Fast and accurate interface.
Administrator	Maintain institution-wide records.	Manage student/course data and reports.	Large volume of records.	Centralized, secure and reliable system.

The supplied brief asks for personas for students, faculty and administrators, including goals, responsibilities, challenges and expectations. filecite turn0file0 L29-L31

10. Requirement Gathering

10.1 Functional Requirements (FRS)

FR1: The system shall allow authorized users to register and maintain student records.

FR2: The system shall maintain course/subject registration information.

FR3: Faculty shall be able to record and update attendance.

FR4: Authorized staff shall be able to maintain examination records.

FR5: Authorized staff shall be able to enter and generate results.

FR6: Students shall be able to view permitted academic information.

FR7: The system shall generate academic reports.

FR8: The system shall provide role-based access to appropriate functions.

These requirements cover the brief's examples of student registration, attendance management, result generation and report generation. filecite turn0file0 L32-L35

10.2 Non-Functional Requirements (NFRS)

Requirement	Description
Security	Only authorized users can access protected academic information.
Usability	Screens should be simple and understandable for intended users.
Reliability	Records and reports should be available and consistent.
Scalability	The system should support growth in students, courses and records.

Security, usability, reliability and scalability are the non-functional requirements named in the supplied brief. filecite turn0file0 L33-L35

11. Product Vision

Vision Statement: To provide a centralized student academic record platform that helps students, faculty and academic administrators manage and access academic information efficiently.

Mission Statement: Build a secure, usable and reliable academic information platform that reduces dependence on fragmented/manual record handling and supports accurate academic reporting.

Success Criteria

Student records can be maintained in one organized system.

Attendance and result information can be recorded and retrieved efficiently.

Academic reports can be generated from maintained records.

Access to information is controlled according to user role.

The prototype receives useful feedback from intended users.

The brief asks for a vision statement, mission statement, success criteria and business objectives. filecite turn0file0 L36-L38

12. Minimum Viable Product (MVP)

The first release should contain the core features required to demonstrate the platform:

- Student registration
- Student record management
- Attendance management
- Academic report generation

These four areas are explicitly identified as the first-release MVP in the supplied brief. filecite turn0file0 L39-L41

13.Product Roadmap

Phase	Work	Expected Output
Phase 1	Business understanding + stakeholder study	Business Requirements Document
Phase 2	Design Thinking + personas	Design Thinking and Persona Reports
Phase 3	Requirement definition	FRS and NFRS
Phase 4	Product vision + MVP	Vision Document and MVP Definition
Phase 5	Prototype + architecture	Prototype and Initial Architecture
Phase 6	Project setup	Git repository and folder structure
Phase 7	Agile development	Working modules and tested releases

The brief requires a phased roadmap aligned with course progression. Filecite turn0file0 L42-L47

14.Project Chater

Item	Details
Project Name	Enterprise Student Record and Academic Management Platform
Purpose	Centralized management of student academic information
Primary Users	Students, faculty and academic administrators
Core MVP	Registration, record management, attendance and academic reports
Constraints	Time, available resources, data privacy and user adoption
Success Measure	Useful, secure, usable and reliable academic-record workflow

15. Project Folder Structure

A simple repository structure can be prepared before development. The exact programming language/framework is not specified in the supplied brief, so the structure below is a planning example.

```
student-academic-platform/  
  
├─ docs/  
|   ├── BRD.md  
|   ├── product-vision.md  
|   ├── stakeholder-analysis.md  
|   ├── personas.md  
|   ├── FRS.md  
|   ├── NFRS.md  
|   ├── design-thinking.md  
|   ├── IBM-design-thinking.md  
|   ├── MVP.md  
|   └─ roadmap.md  
├─ src/  
└─ tests/
```


└─ database/

└─ prototype/

└─ README.md

└─ .gitignore

16. Deliverables Checklist

- ☐ Business Requirements Document (BRD)
- ☐ Product Vision Document
- ☐ Stakeholder Analysis Report
- ☐ User Persona Report
- ☐ Functional Requirements Specification (FRS)
- ☐ Non-Functional Requirements Specification (NFRS)
- ☐ Design Thinking Report
- ☐ IBM Design Thinking Report
- ☐ MVP Definition
- ☐ Product Roadmap
- ☐ Initial System Architecture Diagram
- ☐ Project Charter
- ☐ Project Repository Initialization
- ☐ Project Folder Structure
- ☐ Product Discovery Documentation

This checklist follows the student deliverables listed in the supplied Unit-1 brief. filecite turn0file0 L48-L66.